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BENDABLE DISPLAY DEVICE

Abstract

A display device includes a substrate having a first area, a second area, and a bending area disposed between the first area and the second area. An inner wiring is disposed in the first area. An outer wiring is disposed in the second area. An interlayer insulating layer covers the inner wiring and the outer wiring, and includes a first contact hole. A conductive layer is disposed on the interlayer insulating layer, and is connected to the inner wiring or the outer wiring through the first contact hole. An inorganic protective layer covers at least a portion of the conductive layer and includes an inorganic insulating material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of co-pending U.S. patent application Ser. No. 18/485,289, filed on Oct. 11, 2023, which is a Continuation of U.S. patent application Ser. No. 17/457,964, filed on Dec. 7, 2021 (issued on Oct. 31, 2023 as U.S. Pat. No. 11,805,688), which is a Continuation of U.S. patent application Ser. No. 16/781,555, filed on Feb. 4, 2020 (issued on Dec. 7, 2021 as U.S. Pat. No. 11,195,901), which is a Continuation of U.S. patent application Ser. No. 15/871,631, filed on Jan. 15, 2018 (issued on Feb. 4, 2020 as U.S. Pat. No. 10,551,877), which claims the benefit of Korean Patent Application No. 10-2017-0099784, filed on Aug. 7, 2017, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a display device, and more specifically to a bendable display device.

DISCUSSION OF THE RELATED ART

[0003] A display device may be used in a wide variety of devices. Modern flat panel display devices, such as liquid crystal displays (LCDs) and organic light emitting diode (OLED) displays tend to be thin and lightweight, as compared to older styles of display devices. More recently, many flat panel display devices have the ability to be bent at some point during manufacturing so as to obtain a desired shape.

SUMMARY

[0004] A display device includes a substrate having a first area, a second area, and a bending area disposed between the first area and the second area. An inner wiring is disposed in the first area. An outer wiring is disposed in the second area. An interlayer insulating layer covers the inner wiring and the outer wiring, and includes a first contact hole. A conductive layer is disposed on the interlayer insulating layer, and is connected to the inner wiring or the outer wiring through the first contact hole. An inorganic protective layer covers at least a portion of the conductive layer and includes an inorganic insulating material.

[0005] A display device includes a display area configured to display an image. The display area includes a plurality of pixels. A non-display area is adjacent to the display area. The non-display area includes a bending area. A wiring is disposed in the non-display area. An interlayer insulating layer includes a first contact hole corresponding to the wiring and includes an opening corresponding to the bending area. A conductive layer is disposed on the interlayer insulating layer, and is connected to the wiring through the first contact hole. An inorganic protective layer is

disposed on the conductive layer, and covers at least a portion of the conductive layer corresponding to the first contact hole. The inorganic protective layer includes an inorganic insulating material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] A more complete appreciation of the present disclosure and many of the attendant aspects thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:

[0007] FIG. 1 is a plan view illustrating a display device according to an exemplary embodiment of the present invention;

[0008] FIG. 2 is a perspective view illustrating a portion of a display device according to an exemplary embodiment of the present invention;

[0009] FIG. 3 is a cross-sectional view illustrating a display device according to an exemplary embodiment of the present invention;

[0010] FIG. 4 is a cross-sectional view illustrating a display device according to an exemplary embodiment of the present invention;

[0011] FIG. 5 is a plan view illustrating a portion of a wiring unit of a display device according to an exemplary embodiment of the present invention;

[0012] FIG. 6 is a cross-sectional view illustrating the wiring unit taken along a line VI-VI' of FIG. 5;

[0013] FIG. 7 is a cross-sectional view illustrating the wiring unit taken along a line VII-VII' of FIG. 5;

[0014] FIG. 8 is a plan view illustrating a portion of a wiring unit of a display device according to an exemplary embodiment of the present invention;

[0015] FIG. 9 is a cross-sectional view illustrating the wiring unit taken along a line IX-IX' of FIG. 8;

[0016] FIG. 10 is a cross-sectional view illustrating a display device according to an exemplary embodiment of the present invention;

[0017] FIG. 11 is a cross-sectional view illustrating a display device according to an exemplary embodiment of the present invention;

[0018] FIG. 12 is a plan view illustrating a portion of a wiring unit of a display device according to an exemplary embodiment of the present invention;

[0019] FIG. 13 is a cross-sectional view illustrating the wiring unit taken along a line XIII-XIII' of FIG. 12;

[0020] FIG. 14 is a cross-sectional view illustrating the wiring unit taken along a line XIV-XIV' of FIG. 12;

[0021] FIG. 15 is a cross-sectional view illustrating the wiring unit taken along a line XV-XV' of FIG. 12;

[0022] FIG. 16 is a cross-sectional view illustrating a non-display area of a display device according to an exemplary embodiment of the present invention;

[0023] FIG. 17 is a cross-sectional view illustrating a non-display area of a display device according to an exemplary embodiment of the present invention; and

[0024] FIG. 18 is a plan view illustrating a wiring unit of a display device according to an exemplary embodiment of the present invention.

DETAILED DESCRIPTION

[0025] In describing exemplary embodiments of the present disclosure illustrated in the drawings,

specific terminology is employed for sake of clarity. However, the present disclosure is not intended to be limited to the specific terminology so selected, and it is to be understood that each specific element includes all technical equivalents which operate in a similar manner.

[0026] Like reference numerals in the specification and the drawings may denote like or corresponding elements, and to the extent that repeated description thereof is omitted, it may be assumed that the omitted description is at least similar to descriptions of corresponding elements found elsewhere in the specification.

[0027] It will be understood that although the terms “first”, “second”, etc. may be used herein to describe various components, these components are only used to distinguish one component from another.

[0028] It will be understood that when a layer, region, or component is referred to as being “formed on,” or “disposed on” another layer, region, or component, it can be directly on or intervening layers, regions, or components may be present.

[0029] Sizes of elements in the drawings may be exaggerated for convenience of explanation.

[0030] It will be understood that when a layer, region, or component is referred to as being “connected” to another layer, region, or component, it may be directly connected to the other layer, region, or there may be other layers, regions, or components interposed therebetween.

[0031] A display device is an apparatus displaying an image. The display device may be a liquid crystal display, an electrophoretic display, an organic light-emitting display, an inorganic light-emitting display, a field emission display, a surface-conduction electron-emitter display, a plasma display, a cathode ray display, etc.

[0032] Hereinafter, though an organic light-emitting display is exemplarily described as a display device, the display device according to the present disclosure is not limited thereto and various types of display devices may be used.

[0033] FIG. 1 is a plan view of a display device according to an exemplary embodiment of the preset invention, and FIG. 2 is a perspective view of a portion of a display device according to an exemplary embodiment of the preset invention.

[0034] Referring to FIGS. 1 and 2, a substrate **100** may include a bending area BA extending in a first direction (a Y-direction). The bending area BA may be disposed between a first area **1A** and a second area **2A** along a second direction (an X-direction) crossing the first direction.

[0035] The substrate **100** may be bent about a bending axis BAX extending in the first direction. Though FIG. 2 illustrates that the substrate **100** is bent at the same curvature radius around the bending axis BAX, the invention is not limited thereto. The substrate **100** may be bent around the bending axis BAX and a curvature radius thereof may be varied.

[0036] The substrate **100** may include various flexible or bendable materials, e.g., polymer resins such as polyethersulphone (PES), polyacrylate (PAR), polyetherimide (PEI), polyethylene naphthalate (PEN), polyethylene terephthalate (PET), polyphenylene sulfide (PPS), polyarylate (PAR), polyimide (PI), polycarbonate (PC), and/or cellulose acetate propionate (CAP). The substrate **100** may have a single-layered or multi-layered structure including the above materials. The multi-layered structure may further include an inorganic layer.

[0037] The first area **1A** may include a display area DA. As illustrated in FIG. 1, the first area **1A** may include the display area DA and a portion of a non-display area NDA outside the display area DA. The second area **2A** and the bending area BA may include the non-display area NDA. The display area DA of the display device may correspond to a portion of the first area **1A**, and the non-display area NDA may correspond to the rest of the first area **1A**, the second area **2A**, and the bending area BA.

[0038] The display area DA includes pixels P and may display an image. A pixel P may be connected to signal lines such as a scan line SL extending in the first direction and a data line DL extending in the second direction. In FIG. 1, a pixel P may be connected to power lines transferring a direct-current (DC) signal such as a driving power line and a common power line.

[0039] A pixel P may include electronic elements such as a thin film transistor (TFT) and a storage capacitor electrically connected to the signal lines and power lines, and an organic light-emitting device (OLED) connected to the electronic elements. A pixel P may emit, for example, red, green, blue, or white light through the OLED. The pixel P may be understood herein as being a pixel emitting red, green, or blue light, or a pixel emitting red, green, blue, or white light. The display area DA may be covered by an encapsulation layer **400** so as to protect the display area DA from external air or moisture. In the case where a display element provided to a pixel P is an OLED, the TFT may include a driving TFT and a switching TFT, and may further include an additional TFT in addition to the two TFTs depending on design of the pixel P.

[0040] The non-display area NDA may include first and second scan drivers **11** and **12**, a terminal unit **20**, a driving voltage supply line **30**, a common voltage supply line **40**, and a wiring unit **50**.

[0041] The first and second scan drivers **11** and **12** may be disposed in the first area **1A**. For example, the first and second scan drivers **11** and **12** may be spaced apart from each other with the display area DA therebetween. The first and second scan drivers **11** and **12** may generate and transfer a scan signal to each pixel P through the scan line SL. In FIG. **1**, two scan drivers are disposed, however, the present invention is not limited to using exactly two scan drivers. For example, one scan driver may be disposed on one side of the display area DA.

[0042] The terminal unit **20** may be disposed at one end of the non-display area NDA and includes terminals **21**, **22**, **23**, and **24**. The terminal unit **20** might not be exposed by an insulating layer and be exposed and connected to a flexible film **60** such as a flexible printed circuit board including a driver integrated circuit (IC) **70**. FIG. **1** illustrates a chip on film (COF) type driver IC in which the driver IC **70** is connected through the flexible film **60**, however, the present invention is not limited thereto and other configurations may be used. For example, the driver IC **70** may be a chip on panel (COP) type driver IC in which the driver IC **70** is directly disposed on the terminal unit **20** of the substrate **100**.

[0043] The driving voltage supply line **30** may provide a driving voltage ELVDD to pixels P. The driving voltage supply line **30** may be disposed in the non-display area NDA such that the driving voltage supply line **30** is adjacent to one side of the display area DA.

[0044] The common voltage supply line **40** may provide a common voltage ELVSS to pixels P.

[0045] The common voltage supply line **40** may be disposed in the non-display area NDA to partially surround the display area DA.

[0046] The wiring unit **50** may include inner wirings **210** disposed in the first area **1A**, outer wirings **220** disposed in the second area **2A**, and conductive layers **230**, which are bridge wirings electrically connecting the inner wirings **210** to the outer wirings **220**. Each inner wiring **210** may be electrically connected to a signal line of the display area DA, and each outer wiring **220** may be electrically connected to the terminal unit **20** of the non-display area NDA.

[0047] The conductive layer **230** may extend from the first area **1A** to the second area **2A** across the bending area BA. The conductive layer **230** may cross the bending axis BAX described with reference to FIG. **2**. FIG. **3** illustrates that the conductive layer **230**, which is a bridge wiring, extends perpendicularly with respect to the bending axis BAX, however, the present invention is not limited to this particular arrangement. For example, the conductive layer **230** may obliquely extend at a predetermined angle with respect to the bending axis BAX, or extend while having a curved shape, a zigzag shape, etc., rather than in a straight line shape.

[0048] An organic layer **160** may be disposed in at least the bending area BA. The organic layer **160** at least partially overlaps the conductive layer **230**, one side of the organic layer **160** may extend over the first area **1A**, and the other side of the organic layer **160** may extend over the second area **2A**. In an exemplary embodiment of the present invention, the organic layer **160** may be disposed below the conductive layer **230**, for example, disposed between the substrate **100** and the conductive layer **230**, and may surround a contact portion between the conductive layer **230** and the inner wiring **210**, and a contact portion between the conductive layer **230** and the outer wiring

[0049] FIG. 3 is a cross-sectional view of the display device according to an exemplary embodiment of the present invention, taken along lines A-A' and B-B' of FIG. 1.

[0050] Referring to FIG. 3, an OLED **300** as a display element may be disposed in the display area DA. The OLED **300** may be electrically connected to first and second TFTs T1 and T2 and a storage capacitor Cst. The first TFT T1 includes a first semiconductor layer Act1 and a first gate electrode G1. The second TFT T2 includes a second semiconductor layer Act2 and a second gate electrode G2.

[0051] The first semiconductor layer Act1 and the second semiconductor layer Act2 may each include amorphous silicon, polycrystalline silicon, an oxide semiconductor, or an organic semiconductor material. The first semiconductor layer Act1 includes a channel region C1, and a source region S1 and a drain region D1 respectively disposed at opposite sides of the channel region C1. The second semiconductor layer Act2 includes a channel region C2, and a source region S2 and a drain region D2 respectively disposed at opposite sides of the channel region C2. The source regions S1 and S2 and the drain regions D1 and D2 of the first and second semiconductor layers Act1 and Act2 may be respectively understood as source electrodes and drain electrodes of the first and second TFTs T1 and T2.

[0052] The first gate electrode G1 and the second gate electrode G2 may respectively at least partially overlap the channel region C1 of the first semiconductor layer Act1 and the channel region C2 of the second semiconductor layer Act2 with a gate insulating layer **120** therebetween. The first and second gate electrodes G1 and G2 may each be a single layer or may each have multiple layers including a conductive material including at least one of Mo, Al, Cu, and Ti. FIG. 3 illustrates that the first and second gate electrodes G1 and G2 are disposed on the same layer, however, the present invention is not limited thereto. For example, the first and second gate electrodes G1 and G2 may be respectively disposed on different layers. FIG. 3 illustrates top-gate type TFTs in which the first and second gate electrodes G1 and G2 are respectively disposed over the first and second semiconductor layers Act1 and Act2, however, the present invention is not limited to this particular arrangement. For example, TFTs may be bottom-gate type TFTs in which the first and second gate electrodes G1 and G2 are disposed below the first and second semiconductor layers Act1 and Act2.

[0053] The storage capacitor Cst may include a first storage capacitor plate CE1 and a second storage capacitor plate CE2 which at least partially overlap each other. The first and second storage capacitor plates CE1 and CE2 may include a low-resistance conductive material including at least one of Mo, Al, Cu, and Ti.

[0054] The storage capacitor Cst may at least partially overlap the first TFT T1, and the first TFT T1 may be a driving TFT. Though FIG. 3 illustrates that the storage capacitor Cst at least partially overlaps the first TFT T1 and so the first storage capacitor plate CE1 serves as the first gate electrode G1 of the first TFT T1, the embodiment is not limited thereto. In another embodiment, the storage capacitor Cst might not overlap the first TFT T1.

[0055] A buffer layer **110** may be disposed between the substrate **100** and the first and second TFTs T1 and T2. The buffer layer **110** may include an inorganic insulating material. For example, the buffer layer **110** may be a single layer or multiple layers including at least one of SiON, SiOx, and SiNx.

[0056] The gate insulating layer **120** may be disposed between the first and second gate electrodes G1 and G2 and the first and second semiconductor layers Act1 and Act2. The gate insulating layer **120** may include an inorganic insulating layer. For example, the gate insulating layer **120** may be a single layer or multiple layers including at least one of SiON, SiOx, and SiNx.

[0057] The first and second TFTs T1 and T2 may be covered by an interlayer insulating layer **130**. FIG. 3 illustrates that the interlayer insulating layer **130** includes first and second interlayer insulating layers **131** and **132**. The first interlayer insulating layer **131** may be disposed on the first and second TFTs T1 and T2 and/or on the first storage capacitor plate CE1. The second interlayer

insulating layer **132** may be disposed on the second storage capacitor plate CE2. Each of the first and second interlayer insulating layers **131** and **132** may be a single layer or multiple layers including at least one of SiON, SiOx, and SiNx. In an exemplary embodiment of the present invention, the first interlayer insulating layer **131** may be a single layer including SiNx, and the second interlayer insulating layer may be multiple layers including SiNx and SiOx. The interlayer insulating layer **130** is described herein as being an insulating layer disposed between the inner wiring **210** and the conductive layer **230** and/or the outer wiring **220** and the conductive layer **230**, and may be understood to represent the first interlayer insulating layer **131** or represent the second interlayer insulating layer **132**, or represent the first and second interlayer insulating layers **131** and **132**.

[0058] The data line DL may be disposed on the interlayer insulating layer **130**. The data line DL may be electrically connected to a switching TFT and may provide a data signal. The data line DL may be a single layer or multiple layers including at least one of Al, Cu, Ti, and an alloy thereof. In an embodiment, the data line DL may be three layers including Ti/Al/Ti.

[0059] The data line DL may be covered by an inorganic protective layer PVX. The inorganic protective layer PVX may be a single layer or multiple layers including SiNx and SiOx. The inorganic protective layer PVX may cover and protect some of wirings exposed from the non-display area NDA. Wirings formed simultaneously with the data line DL during the same process may be exposed at a portion of the substrate **100** (for example, a portion of the non-display area NDA). The exposed portion of the wirings may be damaged by an etchant used when a pixel electrode **310** is patterned. Since the inorganic protective layer PVX covers the data line DL and at least a portion of the wirings simultaneously formed with the data line DL, the wirings may be prevented from being damaged during a patterning process of the pixel electrode **310**.

[0060] A driving voltage line PL and the data line DL may be disposed in different layers, for example, with one layer being above the other layer. A first planarization insulating layer **141** may be disposed between the driving voltage line PL and the data line DL.

[0061] The driving voltage line PL may be a single layer or multiple layers including at least one of Al, Cu, Ti, and an alloy thereof. According to an exemplary embodiment of the present invention, the driving voltage line PL may be a triple layer including Ti/Al/Ti. FIG. 3 illustrates that the driving voltage line PL is disposed on only the first planarization insulating layer **141**, however, the present invention is not limited thereto. For example, the driving voltage line PL may be connected to a lower additional voltage line that is formed simultaneously with the data line DL through a through hole formed in the first planarization insulating layer **141**. This arrangement may reduce a resistance as compared to a comparable arrangement without the aforementioned lower additional voltage line connected to the driving voltage line PL via the through hole.

[0062] A second planarization insulating layer **142** may cover the driving voltage line PL. The first and second planarization insulating layers **141** and **142** may include an organic material. The organic material may include a general-purpose polymer such as an imide-based polymer, polymethylmethacrylate (PMMA) or polystyrene (PS), or polymer derivatives having a phenol-based group, an acryl-based polymer, an aryl ether-based polymer, an amide-based polymer, a fluorine-based polymer, a p-xylene-based polymer, a vinyl alcohol-based polymer, or a blend thereof.

[0063] The OLED **300** includes the pixel electrode **310**, an opposite electrode **330**, and an intermediate layer **320** disposed between the pixel electrode **310** and the opposite electrode **330** and including an emission layer. The OLED **300** may be disposed on the second planarization insulating layer **142**.

[0064] A pixel-defining layer **150** may be disposed on the pixel electrode **310**. The pixel-defining layer **150** defines a pixel by having an opening corresponding to each pixel. For example, each opening may expose at least a central portion of the pixel electrode **310**. Also, the pixel-defining layer **150** may prevent arcing, etc. from occurring between the pixel electrode **310** and the opposite

electrode **330** by increasing a distance between the edge of the pixel electrode **310** and the opposite electrode **330**. The pixel-defining layer **150** may include, for example, an organic material such as PI or hexamethyldisiloxane (HMDSO).

[0065] The pixel electrode **310** may be electrically connected to a pixel circuit including, for example, the first and second TFTs T1 and T2 and the storage capacitor Cst through a first connection metal CM1 and a second connection metal CM2.

[0066] The intermediate layer **320** may include a low molecular or polymer material. In the case where the intermediate layer **320** includes a low molecular material, the intermediate layer **320** may have a structure in which a hole injection layer (HIL), a hole transport layer (HTL), an emission layer (EML), an electron transport layer (ETL), an electron injection layer (EIL), etc. are stacked in a single or a composite configuration, and may include various organic materials such as copper phthalocyanine (CuPc), N,N'-Di(naphthalene-1-yl)-N,N'-diphenyl-benzidine (NPB), and tris-8-hydroxyquinoline aluminum (Alq3). These layers may be formed by vacuum evaporation.

[0067] In the case where the intermediate layer **320** includes a polymer material, the intermediate layer **320** may generally have a structure including an HTL and an EML. In this case, the HTL may include PEDOT, and the EML may include a polymer material such as polyphenylene vinylene (PPV)-based material and a polyfluorene-based material. The structure of the intermediate layer **320** is not limited to the above-described structure and may have various structures. For example, the intermediate layer **320** may include a layer having one body over a plurality of pixel electrodes **310** or may include a layer patterned to respectively correspond to the plurality of pixel electrodes **310**.

[0068] The opposite electrode **330** may cover the display area DA. For example, the opposite electrode **330** may have one body over a plurality of OLEDs **300**.

[0069] Since the OLED **300** may be easily damaged by external moisture or oxygen, the OLED **300** may be protected by being covered by an encapsulation layer **400**. The encapsulation layer **400** may cover the display area DA and extend to the outside of the display area DA. The encapsulation layer **400** includes at least one inorganic insulating layer and at least one organic insulating layer. For example, the encapsulation layer **400** may include a first inorganic encapsulation layer **410**, an organic encapsulation layer **420**, and a second inorganic encapsulation layer **430**.

[0070] The first inorganic encapsulation layer **410** may cover the opposite electrode **330** and include SiOx, SiNx, and/or SiON. Other layers such as a capping layer may be disposed between the first inorganic encapsulation layer **410** and the opposite electrode **330**. Since the first inorganic encapsulation layer **410** runs along a structure thereunder, an upper surface of the first inorganic encapsulation layer **410** is not planarized. The organic encapsulation layer **420** covering the first inorganic encapsulation layer **410** may have a flat upper surface at least corresponding to the display area DA. The organic encapsulation layer **420** may include at least one of PET, PEN, PC, PI, polyethylene sulfonate, polyoxymethylene (POM), PAR, and HMDSO. The second inorganic encapsulation layer **430** may cover the organic encapsulation layer **420** and include SiOx, SiNx, and/or SiON.

[0071] An optical film **500** may be disposed on the encapsulation layer **400**. The optical film **500** may include a polarization plate. The polarization plate may reduce external light reflection, and a layer including a black matrix and a color filter may be used instead of the polarization plate. Various functional layers including touch electrode layers may be further provided on the encapsulation layer **400**.

[0072] Referring to the non-display area NDA around the bending area BA of FIG. 3, the buffer layer **110**, the gate insulating layer **120**, and the interlayer insulating layer **130** disposed over the substrate **100** may be collectively referred to as inorganic insulating layer IL. The inorganic insulating layer IL has an opening OP corresponding to the bending area BA. As used herein, the term "correspond" may be understood as overlapping in the vertical direction, as illustrated. The buffer layer **110**, the gate insulating layer **120**, and the interlayer insulating layer **130** may

respectively have an opening **110a**, an opening **120a**, and an opening **130a** corresponding to the bending area BA. The opening **130a** of the interlayer insulating layer **130** may include an opening **131a** and an opening **132a** respectively of the first interlayer insulating layer **131** and the second interlayer insulating layer **132**. Though FIG. 3 illustrates that the opening OP passes through the inorganic insulating layer IL, the present invention is not limited thereto. For example, according to an exemplary embodiment of the present invention, the inorganic insulating layer IL may have a recess (or a groove) corresponding to the bending area BA. The recess may mean a portion formed by removing a portion of the inorganic insulating layer IL in a depth direction and leaving a portion of the inorganic insulating layer IL. For example, the gate insulating layer **120** and the interlayer insulating layer **130** may respectively have the openings **120a** and **130a** corresponding to the bending area BA, but the buffer layer **110** may continuously extend over the first area **1A**, the bending area BA, and the second area **2A** without having an opening.

[0073] The area of the opening OP may be greater than the area of the bending area BA. FIG. 3 illustrates that a width OW of the opening OP is greater than the width of the bending area BA. The area of the opening OP may be defined as the area of a smallest opening among the openings **110a**, **120a**, and **130a** respectively of the buffer layer **110**, the gate insulating layer **120**, and the interlayer insulating layer **130**. FIG. 3 illustrates that the area of the opening OP of the inorganic insulating layer IL is defined by the area of the opening **110a** of the buffer layer **110**. Though FIG. 3 illustrates that an inner surface of the opening **110a** of the buffer layer **110** coincides with an inner surface of the opening **120a** of the gate insulating layer **120**, the present invention is not limited thereto. For example, the area of the opening **130a** of the interlayer insulating layer **130** may be greater than the area of the opening **120a** of the gate insulating layer **120**.

[0074] The organic layer **160** may fill the opening OP of the inorganic insulating layer IL. The organic layer **160** at least partially overlaps the bending area BA, and at least a portion of the organic layer **160** may be disposed inside the opening OP of the inorganic insulating layer IL. The organic layer **160** may extend from the bending area BA to a non-bending area, for example, a portion of the first and second areas **1A** and **2A** adjacent to the bending area BA.

[0075] The organic layer **160** may include at least one of acryl, metacrylic, polyester, polyethylene, polypropylene, PET, PEN, PC, PI, polyethylenesulfonate, polyoxymethylene, PAR, and HMDSO.

[0076] The inner wiring **210** is disposed in the first area **1A** of the non-display area NDA, and the outer wiring **220** is disposed in the second area **2A**.

[0077] The inner wiring **210** may be electrically connected to a pixel of the display area DA. For example, the inner wiring **210** may be electrically connected to the first and second TFTs **T1** and **T2** and/or a wiring such as the data line DL, and the pixel circuit is electrically connected to the display element. The inner wiring **210** may partially extend to the display area DA, or may be electrically connected to a conductive layer/wiring outside the display area DA.

[0078] The outer wiring **220** may be connected to a wiring disposed in a layer different from the layer of the outer wiring **220** in the non-display area NDA and/or the terminal unit **20** (see FIG. 1) of the non-display area NDA. Alternatively, one end of the outer wiring **220** may be exposed to the outside and connected to an electronic device, or the flexible film **60**, etc. described with reference to FIG. 1.

[0079] The inner and outer wirings **210** and **220** spaced apart from each other with the bending area BA therebetween may be connected to the conductive layer **230**. FIG. 3 illustrates a case where the conductive layer **230** is a bridge wiring which crosses the bending area BA and electrically connects the inner wiring **210** to the outer wiring **220**. The conductive layer **230**, which is a bridge wiring, may be disposed in the bending area BA and the first and second areas **1A** and **2A** adjacent to the bending area BA.

[0080] The conductive layer **230** is disposed in a layer different from the layer including the inner and outer wirings **210** and **220** with the interlayer insulating layer **130** therebetween. A portion of the conductive layer **230** corresponding to the bending area BA may be disposed on the organic

layer **160**, and a portion of the conductive layer **230** corresponding to the first and second areas **1A** and **2A** may be disposed in an organic through hole **160h** of the organic layer **160**.

[0081] The organic through hole **160h** may at least partially overlap a first contact hole CNT1 of the interlayer insulating layer **130** and expose the first contact hole CNT1. Opposite ends of the conductive layer **230** are disposed to correspond to the organic through hole **160h** in the first and second areas **1A** and **2A**, and may be connected to the inner and outer wirings **210** and **220** through the first contact hole CNT1. FIG. 3 illustrates a case where two first contact holes CNT1 are disposed inside one organic through hole **160h**, however, the present invention is not limited thereto. For example, number of first contact holes CNT1 provided to the organic through hole **160h** may differ.

[0082] FIG. 3 illustrates a display device in an un-bent configuration, for convenience of description, the substrate **100**, etc. of the display device according to the present disclosure may be bent in the bending area BA as illustrated in FIG. 2. For example, as illustrated in FIG. 3, the display device is manufactured while the substrate **100** is approximately flat, and after that the display device is so-manufactured, the substrate **100** is bent in the bending area BA and may have the shape illustrated in FIG. 2. While the substrate **100**, etc. are bent in the bending area BA, tensile stress may be applied to elements disposed inside the bending area BA.

[0083] To prevent a crack from occurring in the inorganic insulating layer IL due to the aforementioned tensile stress, an opening OP may be formed in a region of the inorganic insulating layer IL corresponding to the bending area BA. As illustrated in FIG. 3, in the case where the conductive layer **230** extends across the bending area BA, a crack may occur in the conductive layer **230** or a defect such as disconnection of the conductive layer **230** may occur during the bending process. To prevent this, the conductive layer **230** may include a material having relatively high elongation. Also, efficiency of electric signal transfer may be increased or a defect occurrence rate during a manufacturing process may be reduced in a display device by forming the inner and outer wirings **210** and **220** of the first and second areas **1A** and **2A** with materials having electric/physical properties different from those of the conductive layer **230**.

[0084] For example, the inner and outer wirings **210** and **220** may include Mo, and the conductive layer **230** may include Al. The inner and outer wirings **210** and **220** and the conductive layer **230** may be a single layer or multiple layers. The conductive layer **230** may be a three-layer configuration of Ti/Al/Ti and may be multiple layers in which a thickness of Ti is 0.15 times the thickness of Al or less, for example, Ti has a thickness of 0.12 times the thickness of Al or less.

[0085] FIG. 3 illustrates a case where the inner and outer wirings **210** and **220** are formed during a process of forming the first and second gate electrodes G1 and G2, and may include the same material as that of the first and second gate electrodes G1 and G2. The conductive layer **230** may be formed during a process of forming the data line DL and may include the same material as that of the data line DL. Therefore, the inorganic protective layer PVX covering the data line DL may cover the conductive layer **230**. In this case, a portion of the inorganic protective layer PVX corresponding to the bending area BA may be removed by an etching process, etc. Therefore, a problem which may occur in the case where the inorganic protective layer PVX remains (for example, a crack may occur in the inorganic protective layer PVX while the substrate **100**, etc. are bent around the bending area BA, or the relevant crack propagates through the inorganic protective layer PVX and damages other layers, etc.) may be prevented or minimized.

[0086] A portion of the inorganic protective layer PVX corresponding to the bending area BA has been removed. The inorganic protective layer PVX may cover at least a portion of the conductive layer **230** in the first and second areas **1A** and **2A**. A distance “d” between an end of the inorganic protective layer PVX disposed in the first area **1A** and an end of the inorganic protective layer PVX disposed in the second area **2A** may be greater than the width of the bending area BA to prevent occurrence and propagation of the crack. The distance “d” between the ends of the inorganic protective layer PVX with the bending area BA therebetween may be substantially equal to, or

greater than, the width OW of the opening OP of the inorganic insulating layer IL. However, the instant invention is not particularly limited to this configuration. For example, the distance “d” may be a little less than the width OW.

[0087] The inorganic protective layer PVX may cover a portion of the conductive layer **230** corresponding to the first contact hole CNT1 in the first area 1A. Referring to the enlarged portion of FIG. 3, the inorganic protective layer PVX may be disposed to correspond to a contact portion “ca” between the inner wiring **210** and the conductive layer **230**, an inner wall **130s** of the interlayer insulating layer **130** surrounding the first contact hole CNT1, and a portion of an upper surface **130u** of the interlayer insulating layer **130** connected to the inner wall **130s**. FIG. 3 illustrates that the inorganic protective layer PVX is disposed on the conductive layer **230** and covers not only a portion of the conductive layer **230** corresponding to the first contact hole CNT1 but also other portions thereof.

[0088] Similarly, the inorganic protective layer PVX may cover a portion of the conductive layer **230** corresponding to the first contact hole CNT1 in the second area 2A. The inorganic protective layer PVX may be disposed to correspond to a contact portion between the outer wiring **220** and the conductive layer **230**, an inner wall of the interlayer insulating layer **130** surrounding the first contact hole CNT1, and a portion of an upper surface of the interlayer insulating layer **130** connected to the inner wall. FIG. 3 illustrates that the inorganic protective layer PVX is disposed on the conductive layer **230** and covers not only a portion of the conductive layer **230** corresponding to the first contact hole CNT1 but also other portions thereof.

[0089] In the case where the inorganic protective layer PVX does not cover at least a portion of the conductive layer **230** and is removed, for example, during an etching process of completely removing the inorganic protective layer PVX in the bending area BA, and the first and second areas 1A and 2A of the non-display area NDA and a subsequent process, damage and/or fine cracks may easily occur on an upper portion of the conductive layer **230**. Damage and/or fine cracks may tend to occur at a step difference around the first contact hole CNT1 which is relatively fragile. Moisture transmission may occur through damage and/or a fine crack, and metal oxide (e.g. AlOx) is generated around the step difference in the vicinity of the first contact hole CNT1 and reduces a resistance of the conductive layer **230**, and the inner and outer wirings **210** and **220** may be disconnected around the metal oxide. However, according to exemplary embodiments of the present invention, since the inorganic protective layer PVX covers at least a portion of the conductive layer **230** corresponding to the first contact hole CNT1, occurrence of the damage and/or fine cracks, moisture transmission, etc. may be prevented.

[0090] A protective layer **600** may be disposed on the conductive layer **230**. Though the first and second planarization insulating layers **141** and **142**, and the pixel-defining layer **150** in the display area DA may extend to the non-display area NDA and form the protective layer **600** covering the conductive layer **230**. However, the invention is not limited to this particular configuration. For example, the protective layer **600** covering the conductive layer **230** in the non-display area NDA may include at least one of the first and second planarization insulating layers **141** and **142** and the pixel-defining layer **150**, or may be formed during a separate process (e.g. coating and curing a material in liquid or paste form) by using an organic material different from those of the first and second planarization insulating layers **141** and **142** and the pixel-defining layer **150**.

[0091] FIG. 4 is a cross-sectional view of a display device according to an exemplary embodiment of the present invention. In FIG. 4, same reference numerals as those of the configuration of FIG. 3 represent the same members. Differences are mainly described below.

[0092] Referring to FIG. 4, the inner and outer wirings **210** and **220** are simultaneously formed during a process of forming the second storage capacitor plate CE2 of the storage capacitor Cst, and may include the same material as that of the second storage capacitor plate CE2. For example, the inner and outer wirings **210** and **220**, and the second storage capacitor plate CE2 may include Mo.

[0093] The inner and outer wirings **210** and **220** may contact the conductive layer **230** via the first contact hole **CNT1**, and at least a portion of the upper surface of the conductive layer **230** may be covered by the inorganic protective layer **PVX** as described above.

[0094] FIGS. **3** and **4** illustrate the driving voltage line **PL** is disposed on the first planarization insulating layer **141** and is covered by the second planarization insulating layer **142**, however, the present invention is not limited to this particular configuration. For example, the driving voltage line **PL** may be formed in the same layer as that of the data line **DL** during the same process. In this case, the second planarization insulating layer **142** may be omitted, and the pixel electrode **310** may be connected to the second TFT **T2** through the first connection metal **CM1**.

[0095] FIGS. **3** and **4** illustrate that the inner wiring **210** and the outer wiring **220** are disposed in the same layer, however, the present invention is not limited thereto. For example, the inner wiring **210** may be disposed on the gate insulating layer **120** as illustrated in FIG. **3**, and the outer wiring **220** may be disposed on the first interlayer insulating layer **131** as illustrated in FIG. **4**.

Alternatively, the inner wiring **210** may be disposed on the first interlayer insulating layer **131** as illustrated in FIG. **4**, and the outer wiring **220** may be disposed on the gate insulating layer **120** as illustrated in FIG. **3**.

[0096] FIG. **5** is a plan view of a portion of a wiring unit of a display device according to an exemplary embodiment of the present invention, and may correspond to a portion **V** of FIG. **1**. FIG. **6** is a cross-sectional view of the wiring unit taken along a line **VI-VI'** of FIG. **5**, and FIG. **7** is a cross-sectional view of the wiring unit taken along a line **VII-VII'** of FIG. **5**.

[0097] Referring to FIG. **5**, the inner wirings **210** may extend in an **X**-direction and may be spaced apart from each other in a **Y**-direction. Hereinafter, for convenience of description, some of the inner wirings **210** is referred to as a first inner wiring **211**, and the rest of the inner wirings **210** is referred to as a second inner wiring **212**. A conductive layer of the conductive layers **230** contacting the first inner wiring **211** is referred to as a first conductive layer **231**, and a conductive layer of the conductive layers **230** contacting the second inner wiring **212** is referred to as a second conductive layer **232**. Also, a through hole of the organic through holes **160h** corresponding to a contact of the first inner wiring **211** and the first conductive layer **231** is referred to as a first organic through hole **160h1**, and a through hole of the organic through holes **160h** corresponding to a contact of the second inner wiring **212** and the second conductive layer **232** is referred to as a second organic through hole **160h2**. As described above with reference to FIGS. **3** and **4**, the first and second conductive layers **231** and **232** illustrated in FIG. **5** may be bridge wirings extending across the bending area **BA** and electrically connecting the inner wiring **210** to the outer wiring **220**.

[0098] A contact region of the first inner wiring **211** and the first conductive layer **231**, and a contact region of the second inner wiring **212** and the second conductive layer **232** may be alternately disposed. For example, the contact region of the first inner wiring **211** and the first conductive layer **231**, and the contact region of the second inner wiring **212** and the second conductive layer **232** may be alternately disposed in a zigzag form. Therefore, a distance between the first inner wiring **211** and the second inner wiring **212** may be reduced and so space may be used more efficiently.

[0099] The organic layer **160** at least partially overlaps the first and second inner wirings **211** and **212** and the first and second conductive layers **231** and **232**, and includes the first and second organic through holes **160h1** and **160h2** for providing contact between the first inner wiring **211** and the first conductive layer **231**, and between the second inner wiring **212** and the second conductive layer **232**. The first organic through hole **160h1** corresponds to an overlapping portion of the first inner wiring **211** and the first conductive layer **231**, and the second organic through hole **160h2** corresponds to an overlapping portion of the second inner wiring **212** and the second conductive layer **232**. The first and second organic through holes **160h1** and **160h2** may be shifted and disposed in a zigzag form.

[0100] The inorganic protective layer **PVX** may correspond to the first and second organic through

holes **160h1** and **160h2**. The inorganic protective layer PVX may cover the contact regions of the first and second inner wirings **211** and **212**, and the first and second conductive layers **231** and **232**. According to an exemplary embodiment of the present disclosure, the inorganic protective layer PVX may partially overlap the organic layer **160**. For example, the inorganic protective layer PVX may overlap at least a portion of the first and second inner wirings **211** and **212** and the first and second conductive layers **231** and **232**.

[0101] Referring to FIGS. **5** and **6**, the interlayer insulating layer **130** including the first contact hole CNT1, for example, the second interlayer insulating layer **132** is disposed on the first inner wiring **211**. The organic layer **160** may be disposed on the interlayer insulating layer **130** and may include the first organic through hole **160h1** formed by removing a portion of the organic layer **160** corresponding to the first contact hole CNT1. A portion of the first conductive layer **231** corresponding to the first organic through hole **160h1** may be connected to the first inner wiring **211** through the first contact hole CNT1.

[0102] Likewise, the second interlayer insulating layer **132** including the first contact hole CNT1 is disposed on the second inner wiring **212**. The organic layer **160** may be disposed on the interlayer insulating layer **130** and may include the second organic through hole **160h2** formed by removing a portion of the organic layer **160** corresponding to the first contact hole CNT1. A portion of the second conductive layer **232** corresponding to the second organic through hole **160h2** may be connected to the second inner wiring **212** through the first contact hole CNT1.

[0103] The first organic through hole **160h1** may be spaced apart by a preset interval from an end of the first inner wiring **211**. For example, a spaced distance between a lower end of the first organic through hole **160h1** and an end of the first inner wiring **211** may be about 4 μm to about 8 μm . Similarly, a lower end of the second organic through hole **160h2** may be spaced apart by the above distance from an end of the second inner wiring **212**.

[0104] The inorganic protective layer PVX may cover a portion of the first conductive layer **231** and a portion of the second conductive layer **232** corresponding to the first contact hole CNT1. Therefore, damage and/or a fine crack occurrence of the first and second conductive layers **231** and **232** and moisture transmission may be prevented as described above with reference to FIG. **3**.

[0105] Referring to FIGS. **5** and **7**, the inorganic protective layer PVX may extend to cover the first conductive layer **231** on the organic layer **160**, and a width W1 in the Y-direction (or a direction crossing an extension direction of the first conductive layer) may be greater than a width W2 of the first conductive layer **231**. Opposite edges of the inorganic protective layer PVX in the Y-direction may contact an upper surface of the organic layer **160**.

[0106] Since the width W1 of the inorganic protective layer PVX is less than a width W3 of the organic layer **160** as illustrated in FIG. **7**, a portion of the upper surface of the organic layer **160** may be exposed to the outside. The portion of the upper surface of the organic layer **160** not covered by the inorganic protective layer PVX may be an out-gassing path through which a material included in the organic layer **160** evaporates and is externally discharged during a curing process of the organic layer **160** in a manufacturing process of a display device.

[0107] FIG. **5** illustrates a case where the first and second inner wirings **211** and **212** are formed during the same process as the second storage capacitor plate CE2 and include the same material as that of the second storage capacitor plate CE2. Here, the first and second conductive layers **231** and **232** are formed during the same process as the data line DL and include the same material as that of the data line DL, as described above with reference to FIG. **4**. However, the present invention is not limited to this particular configuration. For example, the first and second inner wirings **211** and **212** may be formed during the same process as the first and second gate electrodes G1 and G2 and/or the first storage capacitor plate CE1 and include the same material as that of the first and second gate electrodes G1 and G2 and/or the first storage capacitor plate CE1 as described above with reference to FIG. **3**.

[0108] FIGS. **5** to **7** describe a structure in which the inner wiring **210** is connected to the

conductive layer **230**. However, such a structure is equally applicable to a connection structure of the outer wiring **220** and the conductive layer **230** described with reference to FIGS. **3** and **4**. FIG. **8** is a plan view illustrating a portion of a wiring unit of a display device according to an exemplary embodiment of the present invention, and FIG. **9** is a cross-sectional view of the wiring unit taken along a line IX-IX' of FIG. **8**. The arrangement of the first and second inner wirings **211** and **212** and the arrangement of the inorganic protective layer PVX of the display device of FIGS. **8** and **9** are different from that described above with reference to FIGS. **5** to **7**. In FIGS. **8** and **9**, same reference numerals as those of FIG. **5** may represent the same or similar elements. For convenience of description, differences between corresponding elements are mainly described below, and it may be assumed that other details are at least similar to those of corresponding elements that have already been described.

[0109] Referring to FIGS. **8** and **9**, the first and second inner wirings **211** and **212** may be respectively disposed in different layers. For example, the first inner wiring **211** may be disposed between the first interlayer insulating layer **131** and the second interlayer insulating layer **132** and may include the same material as that of the second storage capacitor plate CE2 (see FIGS. **3** and **4**). The second inner wiring **212** may be disposed between the first interlayer insulating layer **131** and the gate insulating layer **120** and may include the same material as that of the first and second gate electrodes G1 and G2 (see FIGS. **3** and **4**) and/or the first storage capacitor plate CE1 (see FIGS. **3** and **4**). In the case where the first and second inner wirings **211** and **212** respectively and alternately disposed in different layers, a distance between the first and second inner wirings **211** and **212** may be further reduced.

[0110] The inorganic protective layer PVX may cover the contact region of the first inner wiring **211** and the first conductive layer **231** and the contact region of the second inner wiring **212** and the second conductive layer **232** and so prevent damage and/or a fine crack from occurring in the first and second conductive layers **231** and **232** at a step difference around the first contact hole CNT1 as described above.

[0111] The inorganic protective layer PVX may at least partially overlap to cover one of the first and second inner wirings **211** and **212**, and may at least partially overlap to cover one of the first and second conductive layers **231** and **232**. FIG. **8** illustrates a case where the inorganic protective layer PVX at least partially overlaps a portion of the first inner wiring **211** and the first conductive layer **231**. A relation between the width W1 of the inorganic protective layer PVX overlapping the first conductive layer **231**, the width W3 of a portion of the organic layer **160**, and the width W2 of the first conductive layer **231** is the same as the relation described above with reference to FIG. **7**.

[0112] The second inner wiring **212** is covered by the first and second interlayer insulating layers **131** and **132** among the inorganic insulating layers IL. On the contrary, the first inner wiring **211** is covered by only the second interlayer insulating layer **132**. To complement a passivation function, which may be relatively insufficient, the inorganic protective layer PVX may at least partially overlap the first inner wiring **211** among the first and second inner wirings **211** and **212** as illustrated in FIG. **8**.

[0113] According to an exemplary embodiment of the present invention, the inorganic protective layer PVX might not overlap the first and second inner wirings **211** and **212**. When the area of the inorganic protective layer PVX covering the organic layer **160** increases, a gas which is not discharged and remains during the curing process of the organic layer **160** may transform the organic layer **160** or layers on the organic layer **160**. Therefore, if a passivation function is sufficiently secured with only the second interlayer insulating layer **132**, the inorganic protective layer PVX may cover only a portion corresponding to contact regions of the first and second inner wirings **211** and **212** and the first and second conductive layers **231** and **232**, and might not cover the first and second inner wirings **211** and **212**.

[0114] FIGS. **8** and **9** illustrate a case where the first inner wiring **211** is disposed between the first interlayer insulating layer **131** and the second interlayer insulating layer **132**, and the second inner

wiring **212** is disposed between the first interlayer insulating layer **131** and the gate insulating layer **120**. However, the present invention is not limited thereto. For example, the first inner wiring **211** may be disposed between the first interlayer insulating layer **131** and the gate insulating layer **120** and may include the same material as that of the first and second gate electrodes **G1** and **G2** and/or the first storage capacitor plate **CE1**. Also, the second inner wiring **212** may be disposed between the first interlayer insulating layer **131** and the second interlayer insulating layer **132** and may include the same material as that of the second storage capacitor plate **CE2** (see FIGS. **3** and **4**).

[0115] Though FIGS. **8** and **9** illustrate a structure in which the inner wiring **210** is connected to the conductive layer **230**, such a structure is equally applicable to a connection structure of the outer wiring **220** and the conductive layer **230** described above with reference to FIGS. **3** and **4**.

[0116] FIG. **10** is a cross-sectional view of a display device according to an exemplary embodiment of the present invention. In FIG. **10**, same reference numerals as those described above with respect to FIG. **3** may represent the same or similar elements. It may be assumed that to the extent that details are not provided, those details may be the same as or similar to details described elsewhere in the disclosure.

[0117] Referring to FIG. **10**, the inner wiring **210** disposed in the first area **1A** may be connected to a conductive layer **230A** through the first contact hole **CNT1** formed in the interlayer insulating layer **130**, and the outer wiring **220** disposed in the second area **2A** may be connected to the conductive layer **230A** through the contact hole **CNT1** formed in the interlayer insulating layer **130**.

[0118] The conductive layer **230A** is disposed in a layer different from the layer in which the inner and outer wirings **210** and **220** are disposed. The inner and outer wirings **210** and **220** are disposed under the interlayer insulating layer **130**, and the conductive layer **230A** may be disposed on the interlayer insulating layer **130**. The inner and outer wirings **210** and **220** may be formed during the same process as the first and second gate electrodes **G1** and **G2** and/or the first storage capacitor plate **CE1**, and may include the same material as that of the first and second gate electrodes **G1** and **G2** and/or the first storage capacitor plate **CE1**. The conductive layer **230A** may be formed during a process of forming the data line **DL** and may include the same material as that of the data line **DL**.

[0119] The conductive layer **230A** is an island type layer and may be disposed in the first and second areas **1A** and **2A**. The conductive layer **230** described with reference to FIG. **3** is a bridge wiring crossing the bending area **BA**, but the conductive layer **230A** of FIG. **10** may correspond to an island electrode layer. The conductive layer **230A** may correspond to the organic through hole **160h**.

[0120] The organic through hole **160h** may at least partially overlap the first contact hole **CNT1** of the interlayer insulating layer **130** and may expose the first contact hole **CNT1**. The conductive layers **230A** may correspond to the organic through holes **160h** in the first and second areas **1A** and **2A**, and may be respectively connected to the inner and outer wirings **210** and **220** through the first contact holes **CNT1**.

[0121] The inorganic protective layer **PVX** may cover a portion of the conductive layer **230A** corresponding to the first contact hole **CNT1**. Referring to an enlarged portion of FIG. **10**, the inorganic protective layer **PVX** may correspond to a contact portion “ca” of the inner wiring **210** and the conductive layer **230A**, the inner wall **130s** of the interlayer insulating layer **130** surrounding the first contact hole **CNT1**, and a portion of the upper surface **130u** of the interlayer insulating layer **130** connected to the inner wall **130s**. A width **W4** of the inorganic protective layer **PVX** overlapping the first contact hole **CNT1** may be greater than a width **W5** of an upper portion of the first contact hole **CNT1**. As described above, since the inorganic protective layer **PVX** covers a step difference around the first contact hole **CNT1**, problems such as occurrence of damage and/or a fine crack in the conductive layer **230A** at the step difference, generation of a metal oxide (e.g. **AlOx**) around the step difference due to generation of moisture transmission, or reduction in a resistance of the conductive layer **230** may be prevented. The inorganic protective layer **PVX**, the first contact hole **CNT1**, and a structure of surroundings thereof illustrated in FIG.

10 are equally applicable to the first contact hole **CNT1** of the outer wiring **220** and surroundings thereof.

[0122] An additional conductive layer **240** may electrically connect the inner wiring **210** of the first area **1A** to the outer wiring **220** of the second area **2A**. The additional conductive layer **240** may be a bridge wiring extending across the bending area **BA** and electrically connecting the inner and outer wirings **210** and **220**.

[0123] The additional conductive layer **240** may include a material having electric/physical properties different from those of the inner and outer wirings **210** and **220**. For example, the inner and outer wirings **210** and **220** may include Mo, and the additional conductive layer **240** may include Al. The additional conductive layer **240** may be a single layer or multiple layers. In an exemplary embodiment of the present invention, the additional conductive layer **240** may be a three-layer arrangement of Ti/Al/Ti and may be multiple layers in which a thickness of Ti is 0.15 times the thickness of Al or less, for example, Ti has a thickness of 0.12 times the thickness of Al or less, but the present invention is not limited to this particular configuration.

[0124] The additional conductive layer **240** may be connected to the conductive layer **230A** through the second contact hole **CNT2** passing through the inorganic protective layer **PVX**. The inner and outer wirings **210** and **220** may respectively contact the conductive layer **230A** through the first contact holes **CNT1** of the interlayer insulating layer **130**, and the additional conductive layer **240** may contact the conductive layer **230A** through the second contact hole **CNT2** of the inorganic protective layer **PVX**, and so the inner and outer wirings **210** and **220** may be electrically connected.

[0125] Since the inorganic protective layer **PVX** should cover a portion of the conductive layer **230A** corresponding to the first contact hole **CNT1** in order to prevent occurrence of damage and/or a fine crack as described above, the second contact hole **CNT2** for providing contact between the conductive layer **230A** and the additional conductive layer **240** may be spaced apart by a preset interval **Ad** such that the second contact hole **CNT2** does not overlap the first contact hole **CNT1**.

[0126] The first planarization insulating layer **141**, which is an organic insulating layer, may be disposed under the additional conductive layer **240**. For example, the first planarization insulating layer **141** of the display area **DA** may extend to the non-display area **NDA** and may be disposed under the additional conductive layer **240**. The first planarization insulating layer **141** may include a hole **141h** at least partially overlapping the second contact hole **CNT2** of the inorganic protective layer **PVX**. FIG. **10** illustrates a case where a plurality of holes **141h** of the first planarization insulating layer **141** are disposed inside the organic through hole **160h** of the organic layer **160**.

[0127] A protective layer **600A** may be disposed on the additional conductive layer **240**. Though the second planarization insulating layer **142** and the pixel-defining layer **150** may extend to the non-display area **NDA** and form the protective layer **600A**, the present invention is not limited to this particular arrangement. For example, the protective layer **600A** covering the additional conductive layer **240** in the non-display area **NDA** may include either the second planarization insulating layer **142** or the pixel-defining layer **150**, or the protective layer **600A** may be formed during a separate process (e.g. coating and curing a material in liquid or paste form) by using an organic material different from those of the second planarization insulating layer **142** and the pixel-defining layer **150**.

[0128] FIG. **11** is a cross-sectional view of a display device according to an exemplary embodiment of the present invention. In FIG. **11**, same reference numerals as those described with reference to FIG. **10** may represent corresponding elements here and any omitted details may be assumed to be at least similar to details of corresponding elements described elsewhere in the specification.

[0129] Referring to FIG. **11**, the inner and outer wirings **210** and **220** are simultaneously formed during a process of forming the second storage capacitor plate **CE2** of the storage capacitor **Cst**, and may include the same material as that of the second storage capacitor plate **CE2**. For example, the inner and outer wirings **210** and **220**, and the second storage capacitor plate **CE2** may include

Mo.

[0130] The inner and outer wirings **210** and **220** may be connected to the conductive layer **230A** through the first contact hole CNT1, at least a portion of the upper surface of the conductive layer **230A** may be covered by the inorganic protective layer PVX, and the additional conductive layer **240** may be connected to the conductive layer **230A** through the second contact hole CNT2 as described above.

[0131] Though FIGS. **10** and **11** illustrate that the inner wiring **210** and the outer wiring **220** are disposed in the same layer, the present invention is not limited to this particular arrangement. For example, the inner wiring **210** may be disposed on the gate insulating layer **120** as illustrated in FIG. **10**, and the outer wiring **220** may be disposed on the first interlayer insulating layer **131** as illustrated in FIG. **11**. Alternatively, the inner wiring **210** may be disposed on the first interlayer insulating layer **131** as illustrated in FIG. **11**, and the outer wiring **220** may be disposed on the gate insulating layer **120** as illustrated in FIG. **10**.

[0132] FIG. **12** is a plan view of a portion of a wiring unit of a display device according to an exemplary embodiment of the present invention, FIG. **13** is a cross-sectional view of the wiring unit taken along a line XIII-XIII' of FIG. **12**, FIG. **14** is a cross-sectional view of the wiring unit taken along a line XIV-XIV' of FIG. **12**, and FIG. **15** is a cross-sectional view of the wiring unit taken along a line XV-XV' of FIG. **12**.

[0133] Referring to FIG. **12**, the first and second inner wirings **211** and **212** may each extend in the X-direction and may be spaced apart from each other in the Y-direction. The first and second inner wirings **211** and **212** may be disposed in the same layer or respectively disposed in different layers. FIG. **12** illustrates a case where the first and second inner wirings **211** and **212** are disposed in different layers.

[0134] The first inner wiring **211** may be connected to the conductive layer **230**, and the second inner wiring **212** may be connected to the conductive layer **230A** and the additional conductive layer **240**. The conductive layer **230** connected to the first inner wiring **211** may be a bridge wiring extending across the bending area BA as described above with reference to FIGS. **3** and **4**. Therefore, the first inner wiring **211** may be electrically connected to the first outer wiring of the second area 2A corresponding to the first inner wiring **211** through the conductive layer **230**.

[0135] The conductive layer **230A** connected to the second inner wiring **212** is an island electrode layer described with reference to FIGS. **10** and **11**. The second inner wiring **212** may be connected to the additional conductive layer **240**, which is a bridge wiring, through the conductive layer **230A**. Therefore, the second inner wiring **212** may be electrically connected to the second outer wiring of the second area 2A corresponding to the second inner wiring **212** through the additional conductive layer **240** by the medium of the conductive layer **230A**.

[0136] Referring to FIG. **13**, the conductive layer **230** and the additional conductive layer **240** may be disposed in different layers. For example, the conductive layer **230** may be formed during the same process as the data line DL (see FIG. **3** or **4**) and may include the same material as that of the data line DL. The additional conductive layer **240** may be formed during the same process as the driving voltage line PL (see FIG. **10** or **11**) and may include the same material as that of the driving voltage line PL. In the case where the conductive layer **230** and the additional conductive layer **240**, which are bridge wirings disposed adjacent to each other, are disposed in different layers, a distance between the conductive layer **230** and the additional conductive layer **240** may be reduced.

[0137] Referring to FIG. **12** again, a contact region of the first inner wiring **211** and the conductive layer **230**, and a contact region of the second inner wiring **212**, the conductive layer **230A**, and the additional conductive layer **240** may be disposed in a zigzag form.

[0138] The organic layer **160** at least partially overlaps the first and second inner wirings **211** and **212** and the conductive layers **230** and **230A** and includes the first and second organic through holes **160h1** and **160h2** for providing contact between the first inner wiring **211** and the conductive layer **230** and between the second inner wiring **212** and the conductive layer **230A**. The first

organic through hole **160h1** corresponds to an overlapping portion of the first inner wiring **211** and the conductive layer **230**, and the second organic through hole **160h2** corresponds to the second inner wiring **212**, the conductive layer **230A**, and the additional conductive layer **240** and may be shifted and disposed in a zigzag form.

[0139] The inorganic protective layer PVX corresponds to the first and second organic through holes **160h1** and **160h2** and covers the contact region of the first inner wiring **211** and the conductive layer **230** and the contact region of the second inner wiring **212** and the additional conductive layer **240**. According to an exemplary embodiment of the present invention, the inorganic protective layer PVX may partially overlap the organic layer **160**. For example, the inorganic protective layer PVX may partially overlap at least one of the first and second inner wirings **211** and **212**, and/or the conductive layer **230**. FIG. 12 illustrates a case where the inorganic protective layer PVX covers the first inner wiring **211** among the first and second inner wirings **211** and **212** and covers the conductive layer **230**. The width W1 of the inorganic protective layer PVX covering a portion of the conductive layer **230** may be greater than the width W2 of the conductive layer **230** and may protect the conductive layer **230**. However, to facilitate out-gassing of a material included in the organic layer **160**, the width W1 may be less than the width W3 of the organic layer **160** as described above.

[0140] As described above, the first inner wiring **211** may be directly connected to the conductive layer **230**, which is a bridge wiring, but the second inner wiring **212** may be connected to the additional conductive layer **240**, which is a bridge wiring, through the conductive layer **230A**. Hereinafter, each connection structure is described.

[0141] First, a connection structure of the first inner wiring **211** and the conductive layer **230** is described.

[0142] Referring to FIGS. 12 and 14, the first inner wiring **211** may be covered by the second interlayer insulating layer **132**, and a portion of the first inner wiring **211** may be exposed through the first contact hole CNT1. The organic layer **160** may be disposed on the interlayer insulating layer **130** and may include the first organic through hole **160h1** formed by removing a portion of the organic layer **160** corresponding to the first contact hole CNT1. The conductive layer **230** may be disposed on the organic layer **160**, and a portion of the conductive layer **230** corresponding to the first organic through hole **160h1** may be connected to the first inner wiring **211** through the first contact hole CNT1. A portion of the conductive layer **230** corresponding to the first organic through hole **160h1** may be connected to the first inner wiring **211** through the first contact hole CNT1. The first organic through hole **160h1** may be spaced apart by a preset distance “d1” from an end of the first inner wiring **211**.

[0143] Since the inorganic protective layer PVX covers a portion of the conductive layer **230** corresponding to the first contact hole CNT1, problems such as occurrence of damage and/or a fine crack, and generation of moisture transmission at surroundings of the first contact hole CNT1, for example, in a step difference at the surroundings of the first contact hole CNT1 may be prevented. The inorganic protective layer PVX is not provided to the bending area BA as described with reference to FIG. 3.

[0144] The first inner wiring **211**, the conductive layer **230**, and a contact region thereof may be covered by the protective layer **600A**. The first planarization insulating layer **141** may be disposed below the protective layer **600A**, for example, between the inorganic protective layer PVX and the protective layer **600A**. The first planarization insulating layer **141** may be disposed under the additional conductive layer **240**, which will be described later with reference to FIG. 15. The first inner wiring **211**, the conductive layer **230**, and a contact region thereof may be entirely covered by the first planarization insulating layer **141**.

[0145] Next, the connection structure of the second inner wiring **212**, the conductive layer **230A**, and the additional conductive layer **240** is described.

[0146] Referring to FIGS. 12 and 15, the second inner wiring **212** may be disposed on the gate

insulating layer **120** and may be connected to the conductive layer **230A** through the first contact hole **CNT1** formed in the interlayer insulating layer **130**.

[0147] The conductive layer **230A** is an island type layer and may be disposed in only a region corresponding to the second organic through hole **160h2**. The area of the second organic through hole **160h2** may be greater than the area of the conductive layer **230A**, and an edge of the conductive layer **230A** may be spaced apart by a preset interval “**d2**” from a lower end of the second organic through hole **160h2**.

[0148] The inorganic protective layer **PVX** may cover a portion of the upper surface of the conductive layer **230A** corresponding to the first contact hole **CNT1** and so prevent occurrence of damage and/or a fine crack, and generation of moisture transmission at a portion of the conductive layer **230A** in a step difference around the first contact hole **CNT1**. The inorganic protective layer **PVX** may at least partially overlap a portion corresponding to the first contact hole **CNT1**, for example, a contact portion of the second inner wiring **212** and the conductive layer **230A**, and an inner wall and a portion of the upper surface of the interlayer insulating layer **130** as described above.

[0149] Since the inorganic protective layer **PVX** covers a portion of the conductive layer **230A** corresponding to the first contact hole **CNT1**, the second contact hole **CNT2** for providing contact between the conductive layer **230A** and the additional conductive layer **240** may be formed in a region different from that of the first contact hole **CNT1**. For example, the second contact hole **CNT2** may be spaced apart from the first contact hole **CNT1**. A lower edge of the second contact hole **CNT2** adjacent to the first contact hole **CNT1** may be spaced apart by an interval “**Ad**” of about 1 μm to 2 μm from an upper edge of the first contact hole **CNT1**. The “**Ad**” may be understood as a spaced distance between the first and second contact holes **CNT1** and **CNT2** adjacent to each other, or as the width of the upper surface of the interlayer insulating layer **130** covered by the inorganic protective layer **PVX** around the first contact hole **CNT1**. The first and second contact holes **CNT1** and **CNT2** may be alternately disposed in the X-direction with the preset interval “**Ad**”.

[0150] The first planarization insulating layer **141** is disposed on the inorganic protective layer **PVX**, and includes the hole **141h** corresponding to the second contact hole **CNT2**. FIG. **15** illustrates a case where the first planarization insulating layer **141** includes a plurality of holes **141h** corresponding to the second organic through hole **160h2**.

[0151] Though FIGS. **12** to **15** illustrate a case where the first inner wiring **211** is disposed on the first interlayer insulating layer **131**, and the second inner wiring **212** is disposed on the gate insulating layer **120**, however, the present invention is not limited to this particular arrangement. For example, the first inner wiring **211** may be disposed on the gate insulating layer **120**, and the second inner wiring **212** may be disposed on the first interlayer insulating layer **131**. According to an exemplary embodiment of the present invention, the first and second inner wirings **211** and **212** may be disposed in the same layer and may include the same material.

[0152] FIGS. **12** to **15** describe a structure in which the inner wiring **210** is connected to the conductive layer **230**, or the conductive layer **230A** and the additional conductive layer **240**. However, such a structure is equally applicable to a structure in which the outer wiring **220** is connected to the conductive layer **230**, or to the conductive layer **230A** and the additional conductive layer **240**.

[0153] FIGS. **16** and **17** are cross-sectional views of a non-display area of a display device according to an exemplary embodiment of the present invention. In FIGS. **16** and **17**, same reference numerals as those of the configuration of FIG. **10** may represent the same or similar elements and to the extent that details are not described, they may be assumed to be at least similar to the details of corresponding elements described elsewhere in the disclosure.

[0154] Referring to FIG. **16**, one hole **141h** may be provided to the first planarization insulating layer **141**. For example, the hole **141h** of the first planarization insulating layer **141** may

correspond to the organic through hole **160h** of the organic layer **160** and may be disposed to expose the second contact hole CNT2. The hole **141h** of the first planarization insulating layer **141** may be less than the organic through hole **160h** and so may provide increased step coverage of the additional conductive layer **240** on the first planarization insulating layer **141**.

[0155] Referring to FIG. **17**, the first planarization insulating layer **141** (see FIG. **10**) might not be disposed under the additional conductive layer **240**.

[0156] FIGS. **16** and **17** illustrate a case where the inner wiring **210** and the outer wiring **220** are disposed in the same layer (e.g. the gate insulating layer), however, the present invention is not limited thereto. The inner wiring **210** and the outer wiring **220** may be disposed on the first interlayer insulating layer **131** as illustrated in FIG. **11**. In an exemplary embodiment of the present invention, the inner wiring **210** and the outer wiring **220** may be respectively disposed in different layers.

[0157] FIG. **18** is a plan view of a wiring unit of a display device according to an exemplary embodiment of the present invention. In FIG. **18**, same reference numerals as those of FIG. **12** may represent the same or similar elements and to the extent that details are not described, they may be assumed to be at least similar to the details of corresponding elements described elsewhere in the disclosure.

[0158] FIG. **12** illustrates a structure in which one of the first and second inner wirings **211** and **212** is connected to the conductive layer **230** serving as a bridge wiring, and the other is connected to the additional conductive layer **240** serving as a bridge wiring by the medium of the island type conductive layer **230A**. Here, both the first and second inner wirings **211** and **212** illustrated in FIG. **18** may be connected to the additional conductive layer **240** by the medium of the conductive layer **230A**.

[0159] Referring to FIG. **18**, the first and second inner wirings **211** and **212** may be respectively connected to the first and second additional conductive layers **241** and **242** serving as bridge wirings by the medium of the island type conductive layer **230A**.

[0160] The first inner wiring **211**, the island type conductive layer **230A**, and the first additional conductive layer **241** at least partially overlap one another in a region corresponding to the first organic through hole **160h1**. The first inner wiring **211** may be connected to the conductive layer **230A** through the first contact hole CNT1 of the interlayer insulating layer, and the conductive layer **230A** may be connected to the first additional conductive layer **241** through the second contact hole CNT2 of the inorganic protective layer PVX.

[0161] Likewise, the second inner wiring **212**, the island type conductive layer **230A**, and the second additional conductive layer **242** at least partially overlap one another in a region corresponding to the second organic through hole **160h2**. The second inner wiring **212** may be connected to the conductive layer **230A** through the first contact hole CNT1 of the interlayer insulating layer, and the conductive layer **230A** may be connected to the second additional conductive layer **242** through the second contact hole CNT2 of the inorganic protective layer PVX.

[0162] The inorganic protective layer PVX may be disposed inside the first and second organic through holes **160h1** and **160h2** and may cover an upper portion of the conductive layer **230A** corresponding to the first contact hole CNT1. In FIG. **18**, the inorganic protective layer PVX may extend to at least partially overlap one of the first and second inner wirings **211** and **212**.

[0163] Since a contact structure of the first inner wiring **211**, the conductive layer **230A**, and the first additional conductive layer **241** is substantially the same as a contact structure of the second inner wiring **211**, the conductive layer **230A**, and the second additional conductive layer **242**, and these structures are the same as those of the embodiment described with reference to FIG. **15**, and to the extent that details are not described, they may be assumed to be at least similar to the details of corresponding elements described elsewhere in the disclosure.

[0164] According to an exemplary embodiment of the present invention, the contact structure of the first inner wiring **211**, the conductive layer **230A**, and the first additional conductive layer **241**, and

the contact structure of the second inner wiring **211**, the conductive layer **230A**, and the second additional conductive layer **242** may be modified to a structure in which one hole **141h** is provided to the first planarization insulating layer **141**, or a structure in which the first planarization insulating layer **141** is omitted in the non-display area NDA as described with reference to FIGS. **16** and **17**.

[0165] FIGS. **3** to **18** illustrate a structure in which the organic layer **160** is disposed under the conductive layers **230** and **230A**, however, the present invention is not limited thereto. If the conductive layer **230** and/or the additional conductive layer **240**, which are bridge wirings in the bending area BA, are flexible, which may endure stress while they are bent, the organic layer **160** may be omitted.

[0166] In describing exemplary embodiments of the present disclosure illustrated in the drawings, specific terminology is employed for sake of clarity. However, the present disclosure is not intended to be limited to the specific terminology so selected, and it is to be understood that each specific element includes all technical equivalents which operate in a similar manner.

Claims

1. A display device, comprising: a substrate comprising a display area and a non-display area, the non-display area including a bending area; a transistor disposed in the display area, the transistor including an active layer and a gate electrode disposed on the active layer; a display element disposed in the display area and electrically connected to the transistor; a first conductive layer disposed in the non-display area; an organic insulating layer disposed in the display area and the non-display area, wherein, in the non-display area, the organic insulating layer is disposed on a top surface of the first conductive layer; a second conductive layer disposed on the organic insulating layer in the non-display area, wherein the second conductive layer is electrically connected to the first conductive layer; and an inorganic protective layer including a first portion that is disposed in the non-display area and a second portion that is disposed in the display area, wherein, in the non-display area, the first portion of the inorganic protective layer is interposed between the top surface of the first conductive layer and a bottom surface of the organic insulating layer, and wherein, in the display area, the second portion of the inorganic protective layer directly contacts the organic insulating layer and an electrode which is electrically connected to the active layer and is interposed between the electrode which is electrically connected to the active layer and the organic insulating layer.
